Leveraging Large Language Models to Derive Multiple Sclerosis Progression Assessments from Clinical Notes: A Feasibility Study

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Introduction

- Problem: Ascertaining Multiple Sclerosis (MS) progression is critical for clinical care and research but is difficult to discern because disability accrual is slow and subtle.
- Standard: The Expanded Disability Status Scale (EDSS) is the gold standard for measuring disability based on eight Functional Systems (FS).
  - Calculating EDSS during routine care is time-consuming and challenging.
  - Details required to infer progression are often locked within unstructured clinical notes.
- Objective: To determine the feasibility of using Large Language Models (LLMs) to automatically extract FS scores and derive EDSS scores from routine clinical notes.

Methods

Cohort Data

- **Source**: University of Pennsylvania Health System Neuroimmunology Registry
- **Sample**: 288 progress notes from 2024-2025 from 239 patients with MS

Annotation

- A single clinician annotator manually reviewed the notes to provide the ground truth labels for both the FS and EDSS scores.

Pipeline

- We utilized OpenAI's GPT-4.1 (Temperature 0.1, Zero-shot) to process notes in two stages:
  - **FS Extraction**: The model extracted scores for 8 Functional Systems: Visual, Brainstem, Pyramidal, Cerebellar, Sensory, Bowel/Bladder, Cerebral, and Ambulation.
  - **EDSS Calculation**: Using Chain-of-Thought reasoning, the model derived the total EDSS score (0.0–10.0) based on the extracted FS scores and specific adjustment rules.

Evaluation

- **FS Metrics**: Accuracy, Weighted/Macro Precision/Recall/F1-scores
- **EDSS Metrics**: Quadratic Weighted Kappa, Mean Absolute Error, Root Mean Squared Error
- **Tolerance Bands**: We assessed "Lenient Matches" (±1.0 point) to account for the "gray zones" in clinical documentation, such as in qualitative descriptions of walking distance, where precise ordinal mapping is difficult and often ambiguous. These bands distinguish "near-misses" from clinically divergent errors, ensuring that reported disagreements reflect only those errors large enough to potentially alter therapeutic decision-making.

Results – FS Scores

FS subtype	Accuracy	Precision	Recall	F1-score	Precision	Recall	F1-score
		Weighted			Macro		
Ambulation	0.799	0.776	0.799	0.784	0.275	0.362	0.295
Bowel/Bladder	0.782	0.876	0.782	0.803	0.476	0.447	0.413
Brainstem	0.900	0.930	0.900	0.911	0.495	0.561	0.511
Cerebellar	0.879	0.892	0.879	0.878	0.691	0.619	0.631
Cerebral	0.787	0.849	0.787	0.801	0.640	0.664	0.641
Visual	0.971	0.970	0.971	0.970	0.708	0.744	0.724
Pyramidal	0.812	0.815	0.812	0.806	0.456	0.436	0.440
Sensory	0.895	0.901	0.895	0.892	0.660	0.638	0.642

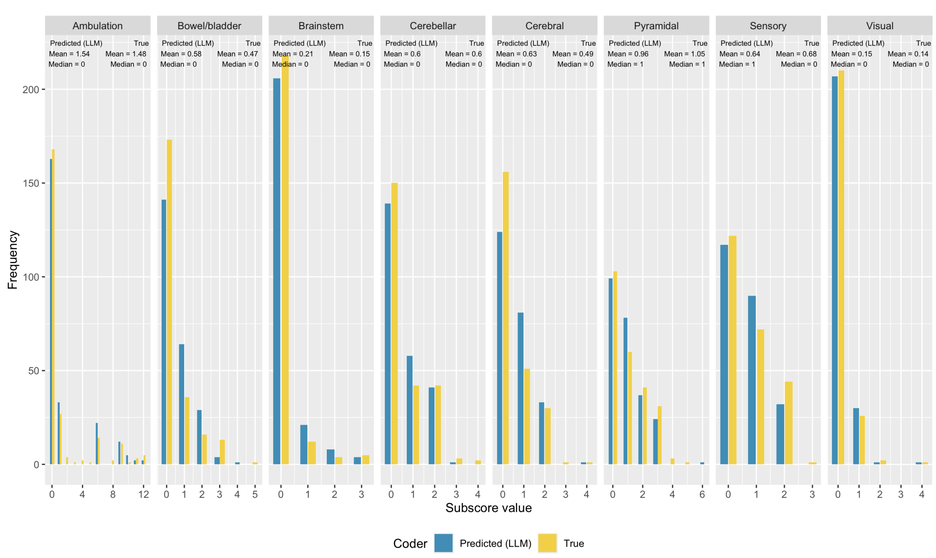

Figure 1.
The model demonstrates strong alignment with expert prevalence rates across domains. While the LLM reliably identifies functional system constructs, it exhibits a slight tendency to over-predict mild impairment in domains like Ambulation and Pyramidal functions, both of which tend to be underspecified in the notes.

Results – EDSS Scores

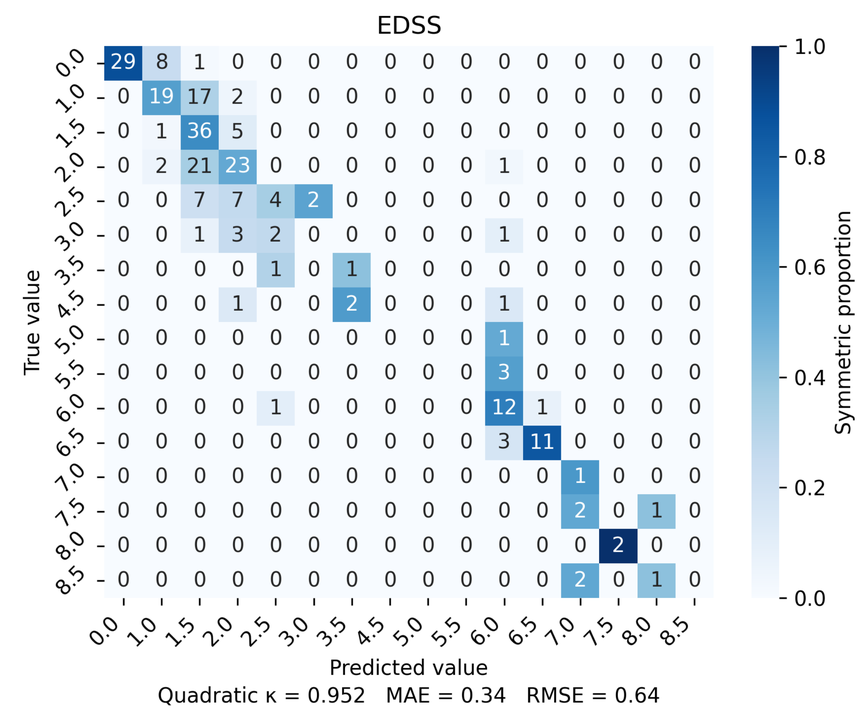

Figure 2.
The strong diagonal density illustrates high agreement, achieving a Quadratic Weighted Kappa of of 0.952. Off-diagonal entries are tightly clustered, indicating that errors are predominantly low-magnitude (off by 0.5) rather than outliers. This suggests the model captures the correct range of disability and that exact ordinal precision may be limited by note ambiguity.

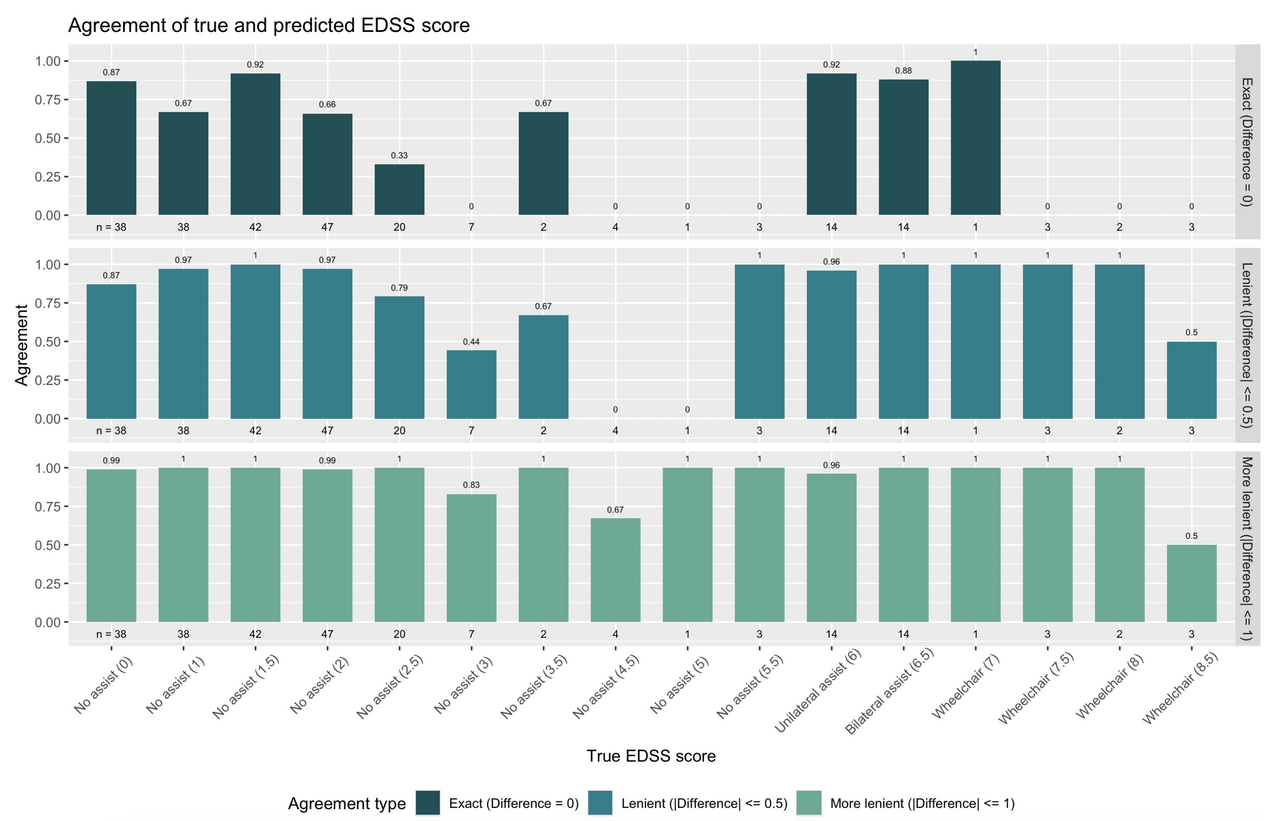

Figure 3.
Agreement is stratified by strictness: **Exact** (dark blue), **Lenient** (±0.5), and **More Lenient** (±1.0). While exact agreement is sensitive to documentation variations, applying a clinically meaningful tolerance band results in high consistency across most scores. This suggests the model is robust enough to track general disability progression trends.

Conclusions

- A single prompt-based LLM can feasibly derive detailed MS progression measures from routine notes.
- While exact EDSS matching is difficult due to underspecified notes, the model performs reliably within clinically meaningful tolerance bands.
- Automating this process can reduce clinician documentation burden and unlock historical data for large-scale outcome studies.

Future Work

- Expand dataset to include multi-site data(Roche-INTONATE) and wide range of MS phenotypes.
- Implement multi-rater annotation to improve ground truth reliability.
- Employ automated prompt optimization to reduce sensitivity to prompt phrasing.
- Develop a "presence-only" formulation to better probabilistically estimate abnormalities without treating "no mention" as negative.

Acknowledgements and References

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- Z. Yang, C. Pou-Prom, A. Jones, M. Banning, D. Dai, M. Mamdani, J. Oh, T. Antoniou et al., Assessment of natural language processing methods for ascertaining the expanded disability status scale score from the electronic health records of patients with multiple sclerosis: algorithm development and validation study, JMIR Medical Informatics 10, p. e25157 (2022).

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